ROHAN VENKATESH

US CITIZEN

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PROFESSIONAL SUMMARY

Aerospace engineering student with experience in propulsion, systems engineering, and computational research spanning CFD, MBSE, hydrogen systems, and defense technology analysis.

EDUCATION

Purdue University, West Lafayette, IN — ***Aug 2023 - May 2027***

Bachelor of Science in Aeronautical and Astronautical Engineering — **GPA: 3.76**

Relevant Coursework - Propulsion, Fluid Dynamics, Thermodynamics, Aerodynamics, Aerospace Design, Dynamics, Structures, Controls, Statistics

TECHNICAL SKILLS

Programming & Analysis - Python, MATLAB, Simulink, C, MySQL
Engineering Software - ANSYS Fluent, Siemens NX, Capella, STK, OpenRocket
Research & Other Tools - Scopus, Ollama/LLMs, Microsoft Excel, xlwings

RELEVANT EXPERIENCE

Space Force Research Intern, Air Force Institute of Technology - Dayton, OH — ***May 2026 - Aug 2026***

- Conducted Space Force-funded research within a 6-intern team, using computational and bibliometric methods to assess technological superiority and identify emerging aerospace and defense technologies.
- Developed MATLAB methods for Applied AI and Scaled Hypersonics, supporting comparison of technology growth, research impact, and emerging technical fields.

Undergraduate Researcher, Li Qiao Group - Purdue University — ***Aug 2025 - May 2026***

- Worked within a 5-student team to develop an autonomous docking and hydrogen-refueling system for NASA-funded "Clean Forever-Flying Drones," supporting long-duration hydrogen UAV operation.
- Converted 3g vertical, 1.5g lateral, and 25 m/s gust conditions into an approximately 3.6 kN docking design load and evaluated latching concepts through weighted trade studies.
- Compared concepts across load capacity, reliability, integration, hydrogen compatibility, and mass, supporting selection of a lightweight push-lock / ball-lock pin architecture.

MBSE Intern, BlueKei Solutions - Pune, India — ***Jul 2025 - Aug 2025***

- Supported MBSE development of reusable launch vehicle landing gear through requirements, functional architecture, verification planning, and failure analysis.
- Developed an AI-enabled Excel/xlwings tool for requirement-quality analysis, traceable functions, V&V criteria, test recommendations, and FMEA outputs, reducing repetitive manual analysis.

Undergraduate Teaching Assistant, Purdue University — ***Aug 2024 - May 2025***

- Led Physics 172 and 272 laboratory instruction for 60+ students across two lab sections, guiding application of mechanics, electromagnetism, and Python-based computational physics.

PERSONAL PROJECTS

Pressure-Fed LOX/CH_4 Rocket Engine Feed System

- Developed a Python model for a 1 kN pressure-fed LOX/CH_4 engine, sizing a 0.363 kg/s propellant flow system and predicting required tank pressures of approximately 26–27 bar.
- Performed parametric studies of feedline diameter, tank pressure, and injector pressure drop to evaluate design sensitivities and pressure margin.
- Built an approximately 75,000-element ANSYS Fluent LOX feedline model, verified close inlet/outlet mass-flow agreement, and compared CFD pressure behavior against the analytical model.

Rocketry - Tripoli Level 1 Certification

- Earned Tripoli Level 1 high-power rocketry certification at Thunderstruck 2024 using a LOC IV rocket and AeroTech H100W motor.
- Used OpenRocket to compare motor configurations and wind conditions and evaluate flight performance, stability, and recovery deployment.